MATHEMATICS

SECTION – I

Single Correct Choice Type

1. (b) Total number of cases = $^{20}C_{10} \times 10! \times 10!$

Favorable cases = $^{10}C_5 \times 10! \times 10!$

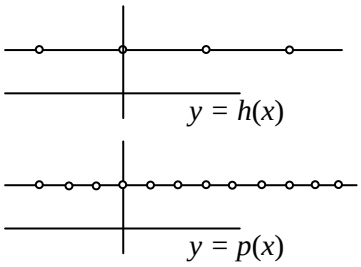
Hence required probability = $\frac{^{10}C_5}{^{20}C_{10}}$

Consider that after each draw, the person arrange the balls in the order.

2. (b) $h(x) = \frac{\{x\}}{\{x\}}$

$p(x) = \frac{g(x)}{g(x)}$

Obviously the combined period will be 1/4 if $g(x)$ becomes zero at equal interval of 1/4 and as well as at the integers.



3. (b) $f'\left(\frac{x}{y}\right).f\left(\frac{y}{x}\right) = \frac{x^2+y^2}{xy}$ …(i)

Interchanging x, y we get

$$f'\left(\frac{x}{y}\right).f\left(\frac{y}{x}\right) = f'\left(\frac{y}{x}\right)f\left(\frac{x}{y}\right) \Rightarrow \ln f\left(\frac{x}{y}\right) = \ln f\left(\frac{y}{x}\right) + C$$

Putting $x = y$, we get $C = 0 \Rightarrow f\left(\frac{x}{y}\right) = f\left(\frac{y}{x}\right)$ …(ii)

Putting $y = 1,$ in equation (i) and (ii), we get

$\Rightarrow f'(x)f\left(\frac{1}{x}\right) = \frac{x^2+1}{x}$ and $f(x) = f\left(\frac{1}{x}\right)$

$\Rightarrow f'(x)f(x) = x + \frac{1}{x} \Rightarrow (f(x))^2 = x^2 + 2\ln x$

4. (b) $(x^2+2x+4)^n = \sum_{r=0}^{2n} a_r x^r$

Let $x = 2t \Rightarrow 4^n(1+t+t^2)^n = \sum_{r=0}^{2n} 2^r a_r t^r$

Let $x = \frac{2}{t} \Rightarrow 4^n(1+t+t^2)^n = \sum_{r=0}^{2n} 2^r a_r t^{2n-r}$

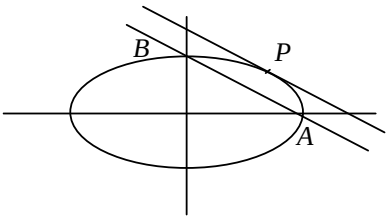
On comparing coefficient of t_r

$\Rightarrow 2^r a_r = 2^{2n-r} a_{2n-r} \Rightarrow \sum_{r=n}^{2n} \frac{a_{2n-r}}{a_r} = \sum_{r=n}^{2n} \frac{1}{4^{n-r}} = \frac{4^{n+1}-1}{3}$

5. **(c)** Let perpendicular distance of P from the line be h

$\frac{1}{2}\times h\times 5 = 6(\sqrt{2}-1)$ (as $\Delta PAB = 6(\sqrt{2}-1)$)

$\Rightarrow \quad h = \frac{12(\sqrt{2}-1)}{5}$



Now distance of tangent parallel to AB i.e. $4y+3x=12\sqrt{2}$, from line AB is $\frac{12(\sqrt{2}-1)}{5}$.

There are just three such points.

6. **(d)** $z = a+ib$

$\Rightarrow \quad |a+ib|^{2010} = |a-ib|$

$\Rightarrow \quad |z|^{2010} = |z| \quad \Rightarrow \quad |z| = 0 \quad \Rightarrow \quad (a, b) \equiv (0, 0)$

or $\quad |z| = 1 \quad \Rightarrow \quad z^{2010} = \frac{1}{z} \quad \Rightarrow \quad z^{2011} = 1$

Hence z has 2012 roots.

SECTION – II
Multiple Correct Choice Type

7. **(a), (c)**

$$f\left(g(n, h(x))\right) = \left|\sin\pi\left([n]+[n]^2\right)^{\frac{x}{\log\left(\frac{e^x+e^{2x}+e^{3x}}{3}\right)}}\right|$$

$$\lim_{x\to 0} f(g(n, h(x)) = \left|\sin\pi([n]+[n]^2)^{\frac{1}{2}}\right|$$

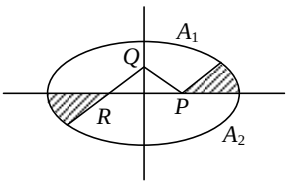
$$\Rightarrow \quad \lim_{n\to\infty}\sin\left(\pi[n]\left(1+\frac{1}{[n]}\right)^{1/2}\right) = \lim_{n\to\infty}\left|\sin\left(\pi[n]+\frac{\pi}{2}-\frac{1}{[n]}\times\frac{\pi}{8}\ldots\right)\right| = 1$$

$$\lim_{n\to\infty}\lim_{x\to 0} g(n,\ h(x)) = \lim_{n\to\infty}\sin\left(\pi[n]+\frac{\pi}{2}-\frac{1}{8[n]}+\ldots\right)$$

does not exists as value can be 1 or –1.

8. **(a), (c)**

Area of ellipse $= \pi\times\frac{8}{3\sqrt{\pi}}\times\frac{5}{2\sqrt{\pi}} = \frac{20}{3} = A_e$



Both shaded areas are equal, then

$A_1 = \frac{1}{2}A_e - \Delta PQR = \frac{10}{3} - 1$

$A_2 = \frac{1}{2}A_e + \Delta PQR = \frac{10}{3} + 1$

9. (a), (b), (c)

If n is even

$$f(n) = 2\left(1+2+\ldots\frac{n-2}{2}\right)+\frac{n}{2} = \frac{n^2}{4}$$

If n is odd then $f(n) = f(n-1)+\frac{n-1}{2} = \frac{(n-1)^2}{4}+\frac{n-1}{2} = \frac{n^2-1}{4}$

$$f(n+m)-f(n-m) = \frac{(n+m)^2-(n-m)^2}{4} \text{ if both even}$$

$$= \frac{(n+m)^2-1-(n-m)^2+1}{4} \text{ it both odd}$$

$$= nm$$

10. (a), (b), (c) & (d)

$$f_1(x) = \frac{\pi}{2}-\sin^2 x,\quad f_2(x) = \frac{\pi}{2}-\cos^2 x$$

$$f_3(x) = \frac{\pi}{2}-\cos^2 x,\quad f_4(x) = \frac{\pi}{2}-\sin^2 x$$

11. (a), (b) & (c)

If a, b, c are odd integer then D can not be perfect square,

$\Rightarrow \quad |f(\lambda)| > 0$ where $\lambda \in Q$

$\Rightarrow \quad |a p^2 + b pq + c q^2| \geq 1 \quad \Rightarrow \quad \left|f\left(\frac{p}{q}\right)\right| \geq \frac{1}{q^2}$

If $a.c = 1 \quad \Rightarrow \quad a = c = 1$ and $b \geq 3$ as

$c.(a-b+c) < 0 \quad \Rightarrow \quad \alpha \in (-1, 0)$

SECTION–III
Reasoning Type

12. (a) $f'(x) = \frac{h(x)g'(x)-h'(x)g(x)}{(h(x))^2} > 0$

(As $h(x), g(x) > 0, g'(x) > 0$ and $h'(x) < 0$)

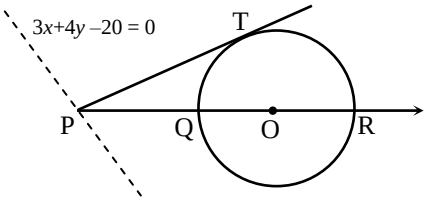
13. (a) Let $OP = x$

$PT^2 = PQ.PR = (OP-1):(OP+1) = (OP)^2-1$

For PT minimum OP should be minimum.

Minimum distance of OP from (0, 0) to $3x+4y-20=0$ is 4

$\Rightarrow \quad PT^2 = 16-1 = 15 \quad \Rightarrow \quad PT = \sqrt{15}$



14. (a) The value of cofactors of elements of particular row remain unchanged if we change a particular row with any other row, which will make two row identical and hence value of change determinant will become zero.

15. (a) $f(x,\ y)\, f\left(\frac{\pi}{2}-x, \frac{\pi}{2}-y\right)$

$$\Rightarrow \quad \left(\frac{1+\sin^2 x}{1+\sin^2 y}-1\right)\left(\frac{1+\cos^2 x}{1+\cos^2 y}-1\right)=\frac{(\sin^2 x-\sin^2 y)(\cos^2 x-\cos^2 y)}{(1+\sin^2 y)(1+\cos^2 y)} \le 0$$

$$(\text{as } (\sin x-\sin y)(\cos x-\cos y)\le 0,\ \ \forall x,\ y\in(0,\pi/2))$$

$\Rightarrow$ Exactly one of $(f(x,\ y)+1), \left(f\left(\frac{\pi}{2}-x, \frac{\pi}{2}-y\right)+1\right)$ is greater than 1.

$\Rightarrow$ Statement 1 is true.

16. (b) Let $T_{r+1} \ge T_r$

$$\Rightarrow \quad \frac{{}^nC_r}{(r+1)(r+2)} \ge \frac{{}^nC_{r-1}}{r.(r+1)}$$

$$r \le \frac{n-1}{2} \ \Rightarrow \ r \le 5.5$$

S0 6th term of sequence greatest.

SECTION–IV
Matrix–Match Type

1. (A) – 2, 4; (B) – 3, 5; (C) – 2, 3, 5; (D) – 1, 2, 4

(A) $S_n = \sum_{r=1}^{n} \cos^4\left(\theta+\frac{2(r-1)\pi}{n}\right)$

$$=\frac{1}{4}\sum_{r=1}^{n}\left(1+\cos\left(2\theta+\frac{4(r-1)\pi}{n}\right)\right)^2$$

$$=\frac{1}{4}\sum_{r=1}^{n}\left(1+\cos^2\left(2\theta+\frac{4(r-1)\pi}{n}\right)+2\cos\left(2\theta+\frac{4(r-1)\pi}{n}\right)\right)$$

$$=\frac{1}{4}\sum_{r=1}^{n}\left(1+\frac{1}{2}\left(1+\cos\left(4\theta+\frac{8(r-1)\pi}{n}\right)\right)\right)$$

$$=\frac{1}{4}\times\frac{3}{2}n=\frac{3}{8}n$$

(B) $\cos mA+\cos mB+\cos mC=2\cos m\left(\frac{A+B}{2}\right)\cos m\left(\frac{A-B}{2}\right)+1-2\sin^2\frac{mC}{2}$

$=1+2\sin\frac{mC}{2}\left(\cos m\left(\frac{A-B}{2}\right)-\sin\frac{mC}{2}\right)$ for $m=(4n+1)$

$=1+2\sin\frac{mC}{2}\left(\cos m\left(\frac{A-B}{2}\right)-\cos m\left(\frac{A+B}{2}\right)\right)$

$=1+4\sin\frac{mA}{2}\sin\frac{mB}{2}\sin\frac{mC}{2}$

(C) $\cos^n x=1+\sin^n x$, for n even $x=m\pi$,

For n odd $x=2m\pi$ or $x=\left(2p\pi+\frac{3\pi}{2}\right)$

(as $\cos^n x+\sin^n x\le\sin^2 x+\cos^2 x=1$ if $n>2$)

(D) $\sin x=\frac{6}{5m-m^2}$ or $\frac{4}{5m-m^2}$, but for exactly three roots $\sin x=\pm1$

$\Rightarrow \quad \frac{6}{5m-m^2}=\pm1 \quad \Rightarrow \quad m=2,3,-1,6$

2. (A) – 2, 5; (B) – 2, 4; (C) – 2, 3; (D) – 1, 2, 4

(A) $\lim\limits_{n\to\infty}\frac{\left(\sum\limits_{x=1}^{n}x^4\right)\left(\sum\limits_{x=1}^{n}x^5\right)}{\left(\sum\limits_{x=1}^{n}x^t\right)\left(\sum\limits_{x=1}^{n}x^{9-t}\right)}=\frac{\int_0^1 y^4dy\int_0^1 y^5dy}{\int_0^1 y^t dy\int_0^1 y^{9-t}dy}=\frac{(t+1)(10-t)}{30}=\frac{4}{5}$

$\Rightarrow \quad t^2-9t+14=0, \quad (t-2)(t-7)=0, \quad t=2,7$

(B) $x^y=y^x \quad \Rightarrow \quad y\ln x=x\ln y$

Differentiating with respect to to x we get $y'=\frac{(x\ln y-y)}{(y\ln x-x)}=\frac{y^2(\ln x-1)}{x^2(\ln y-1)}$

$y'=\lim\limits_{\substack{x\to e\\y\to e}}\frac{y^2(\ln x-1)}{x^2(\ln y-1)}=\lim\limits_{\substack{x\to e\\y\to e}}\frac{y}{x\ y'}$

$\Rightarrow \quad y'=\pm1$

$\Rightarrow$ So $m=\pm1$ and $3-m$ is equal to 2 or 4

(C) Put $x = y = 1$

$\Rightarrow \quad (f(1))^2 = f(1) + 6 \quad \Rightarrow \quad f(1) = 3, -2$

Put $y = 1$, we get $f(x) = \frac{3}{2}\left(\frac{1+x}{x}\right)$ or $f(x) = -\left(\frac{1+x}{x}\right)$

(D) If m is even $(\sqrt{2}+1)^m + (\sqrt{2}-1)^m = 2I$

$\Rightarrow \quad \lim_{n\to\infty}\{(\sqrt{2}+1)^{n!+k}\} = \lim_{n\to\infty}\{2I - (\sqrt{2}-1)^{n!+k}\} = \lim_{n\to\infty}\{-(\sqrt{2}-1)^{n!+k}\} = 1$

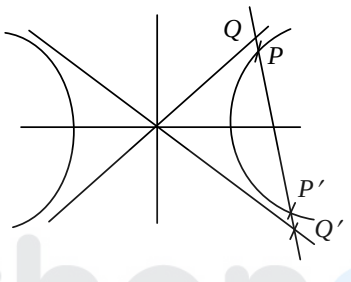
$\Rightarrow \quad n!+k$ must be even integer $\Rightarrow$ k is even.

SECTION–V
Integer Answers Type

1. For any line in case of hyperbola we know that $PQ = P'Q'$

$$\frac{PQ}{P'Q'} + \frac{PQ'}{P'Q} = 1 + \frac{PP' + P'Q'}{PP' + PQ} = 2$$

Ans. 2



2. The number of ways when no student failed in all the examination $= (2^3 - 1)^3$

The number of ways when out of above cases atleast one examination was not cleared by any students $= {}^3C_1(3)^3$

The number of ways when out of above cases atleast any two examination were not cleared by any student $= {}^3C_2(1)^3$

So required cases $= (2^3 - 1)^3 - {}^3C_1 3^3 + {}^3C_2 = 265 = n$

Ans. 5

3. $f(x) = 30 - 2x - x^3$

$f'(x) = -2 - 3x^2 < 0 \quad \Rightarrow \quad f(x)$ is decreasing function

Hence $f(f(f(x))) > f(f(-x)) \quad \Rightarrow \quad f(f(x)) < f(-x)$

$\Rightarrow \quad f(x) > -x$

$\Rightarrow \quad 30 - 2x - x^3 > -x \Rightarrow x^3 + x - 30 < 0 \quad \Rightarrow \quad (x-3)(x^2+3x+10) < 0$

$\Rightarrow \quad x < 3$

Ans. 2

4. $\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}^k = \left(\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}\right)^k = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}+k\begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix} \quad \left(\text{as } \begin{bmatrix}2 & -4\\ 1 & -2\end{bmatrix}^2 = \text{O}\right)$

$$= \begin{bmatrix}2k+1 & -4k\\ k & 1-2k\end{bmatrix}$$

$$\Rightarrow \quad \sum_{k=1}^{n} A^k = \begin{bmatrix}n+n(n+1) & -2n(n+1)\\ \dfrac{n(n+1)}{2} & n-n(n+1)\end{bmatrix} \Rightarrow \quad \det.\left(\sum_{k=1}^{n} A^k\right) = n^2 = 64$$

$$\Rightarrow \quad n = 8$$

Ans. 8